Yeshua Adnelle S. Arjona BSIT 2-4 Object-Oriented Programming

Part A - Variable Identification Table

Variable Name	Data Type	Variable Type	Stored in
name	string	instance	Heap
grade	integer	instance	Heap
n	string	(Local) Parameter	Stack
g	integer	(Local) Parameter	Stack
s1	object	(instance) Reference	Stack
s2	object	(instance) Reference	Stack
s3	object	(instance) Reference	Stack

part B

Statement	What Happens?	Memory Impact
Student s1 = new Student ("John", 85);	A new student object is created with name = John and grade = 85. Reference variable s1 points to it.	A new object is allocated on the heap. s1 stores the memory address of that object.
Student s2 = new Student ("Mana", 90);	A new student object is created with name = Mana and grade = 90. Reference variable s2 points to it.	A second new object is allocated on the heap. s2 stores its address
Student s3 = s2;	s3 is assigned the same reference as s2. No new object is created. Both s3 and s2 now point to the same object	No new heap allocation. s3 stores the same address as s2.
s1.grade = 95;	The grade field of the object referenced by s1 is updated from 85 to 95.	The existing object on the heap is modified. No new object created
s3.name = "Ana";	The name field of the object referenced by s3 is changed to "ANA". Since s3 and s2 points at the same object, Mana's object is Affected	The same heap by s2 and s3 is modified. name changes from Mana to Ana

NO.
DATE

Part c - output

John 95
Ana 90

Ana 90